

CHAPTER 3

RSOAE: AN INTELLIGENT GLAUCOMA PREDICTION SYSTEM FOR DIABETIC MELLITUS COMMUNITY

3.1 OVERVIEW OF THE PROPOSED RESEARCH

Glaucoma is an interfering retinal complaint that points to long-lasting prophecy letdown starved of any earlier sign. To evade conjuring up problems scheduled fact, the inconvenient discovery of glaucoma is noteworthy among the diabetic community. An intellectual forecast method is aimed to identify glaucoma using the ML exemplary. In this recommended method, Random Search Optimization AdaBoost Ensemble (RSAOE) is fused into the diabetic datatable and obtains the forecast of glaucoma by means of the accurateness of 97.5%, 98.01%, and the area under the curve of 0.96, 0.97 on learning and testing set, correspondingly. This model is enhanced than the Bayesian Optimization Support Vector Machine (BOSVM) ideal by way of an inference time of 0.27. This innovative assessment support system can foretell rigorousness among diabetic persons and post-fungal eye contagions later surgery.

The proposed method utilizes bilateral filtering to denoise and enrich the eminence of retinal images. It ensures that the input data is free from noise and artifacts for better model training. It uses methods RF to finest the best significant attributes from clinical information along with image data. RSAOE applied to predict glaucoma risk and outputs predictions with confidence scores for clinical decision-making.

The core objective of RSOAE is to boost early detection besides prediction of glaucoma in the diabetic mellitus population by integrating image- and data-driven approaches using advanced ML and DL techniques. It focuses on localized areas of the retinal images where glaucoma biomarkers are prevalent. The proposed method improves accuracy by prioritizing critical regions, instance of

OD and RNFL. It uses attention mechanisms to assign higher weights to crucial features in the data. RSOAE ensures the system is highly sensitive to early signs of glaucoma. It utilizes multimodal data, including fundus images, IoP, and patient demographics, to improve prediction performance. This system represents a significant advancement in personalized and intelligent glaucoma prediction, particularly for vulnerable populations like those with diabetes.

3.1.1 Contributions of the RSOAE System

- The optimization process refines the prediction model, ensuring high precision and reliability in identifying at-risk individuals.
- The strategic contribution of the RSOAE system is its ability to integrate multiple types of data, including medical imaging (e.g., OCT, fundus images), patient health records, and physiological parameters (e.g., IoP).
- The RSOAE system leverages Swarm Optimization Algorithms (SOA) handpicked the maximum applicable features from large, complex datasets. This feature selection improves the model's efficiency by concentrating on the utmost significant predictors of glaucoma risk, reducing computational overhead and increasing the speed of predictions.
- The RSOAE system can handle sequential data, such as changes in intraocular pressure over time or patient history, which is crucial for predicting the progression of glaucoma.
- The primary benefit of the RSOAE system is ability to detect glaucoma in its initial times, which is critical for preventing irreversible vision damage. The automation of the glaucoma prediction process through the RSOAE system can moderate the job of healthcare authorities by streamlining the diagnostic process.

3.2 PROPOSED METHODOLOGY

3.2.1 RSOAE Method

RSOAE method is a hyperparameter tuning technique that randomly samples hyperparameters from specified distributions to find the best-performing model (Figure 3.1). It is an ensemble learning method that merges multiple weak classifiers (typically decision trees) to form a solid classifier. RSOAE sequentially trains weak models, emphasizing samples that were previously misclassified, thus improving classification accuracy iteratively. It is employed to optimize the hyperparameters of Adaboost (e.g., weak classifiers number, learning rate, and tree depth for weak learners). It selects hyperparameter values randomly from a predefined range, identifying the optimal configuration efficiently. Misclassified images are assigned higher weights, so the subsequent classifier focuses more on these challenging cases. The final classification is grounded on a weighted mainstream vote of all weak classifiers. The RSOAE model, after feature selection, classifies glaucoma images into categories (e.g., glaucomatous or normal). In equation (3.1) we perform the objective function,

$$i = \{i_1, i_2, \dots, i_n\} \quad (3.1)$$

by assuming, i as an hypermeter, the equation(3.1) is used to minimize or maximize a $f(i)$ function. Then we perform the search space through equation(3.2),

$$\theta = \{\theta_1 \in [x_1, y_1], \theta_2 \in [x_2, y_2], \dots, \theta_n \in [x_n, y_n]\} \quad (3.2)$$

Let us assume, θ as parameter space and $[x_a, y_a]$ as the ranges for each hyperparameter. This equation defines the parameter space θ within which the random samples are drawn. By following of what we process the sampling through equation (3.3),

$$\theta^{(b)} \sim U(\theta), \quad \forall b \in \{1, 2, \dots, s\} \quad (3.3)$$

Let assume s as samples, equation (3.3) randomly generates the s . Through equation (3.4), we evaluate the performance metric $\mathcal{M}$, as follows

$$\mathcal{M}(\theta^{(b)}) = f(\theta^{(b)}) \quad (3.4)$$

Let assume each sampled $\theta^{(b)}$ evaluate the $\mathcal{M}$. After measure the $\mathcal{M}$ we select the appropriate parameter through equation (3.5),

$$\theta^* = \arg \max_{\theta^{(b)} \in \Theta} \mathcal{M}(\theta^{(b)}) \quad (3.5)$$

In this above equation when we choose θ^* , it optimizes the $\mathcal{M}$. In equation (3.6) we initialize the weights u , for n training samples, as

$$u_a^{(1)} = \frac{1}{n}, \quad \forall a \in \{1, 2, \dots, n\} \quad (3.6)$$

By the weak classifier $g_z(i)$ are trained at iteration (z) through equation (3.7),

$$g_z(i) = \arg \min_g \sum_{a=1}^n u_a^{(z)} \cdot 1(g(i_a) \neq j_a) \quad (3.7)$$

After training the $g_z(i)$, we compute the weighted classification error through equation (3.8), as

$$\varepsilon_z = \frac{\sum_{a=1}^n u_a^{(z)} \cdot 1(g_z(i_a) \neq j_a)}{\sum_{a=1}^n u_a^{(z)}} \quad (3.8)$$

After computing the classification error we assign the weight to the weak classifier based on its error using the equation (3.9),

$$\alpha_z = \frac{1}{2} \ln \left(\frac{1 - \varepsilon_z}{\varepsilon_z} \right) \quad (3.9)$$

We update and normalize the sample weights for the next iteration through the equation (3.10) and (3.11),

$$u_a^{(z+1)} = u_a^{(z)} \cdot \exp(-\alpha_z \cdot j_a \cdot g_z(i_a)) \quad (3.10)$$

We perform normalize at below equation (3.11) at below,

$$u_a^{(z+1)} = \frac{u_a^{(z+1)}}{\sum_{y=1}^n u_a^{(z+1)}} \quad (3.11)$$

In equation (3.12), we combine all weak classifiers to form the final model,

$$G(a) = \text{sign} \left(\sum_{z=1}^Z \alpha_z \cdot g_z(a) \right) \quad (3.12)$$

The proposed method is used to generate candidate hyperparameter values. This framework effectively classifies glaucoma images by leveraging the strength

of Adaboost and the flexibility of Random Search Optimization (RSO) effectively identifies the optimal hyperparameters for Adaboost, leading to improved model performance compared to manual tuning or grid search techniques. The deployed methods used to combine weak classifiers into a strong ensemble, coupled with optimized parameters, results in robust classification of glaucoma images. The proposed method ensures a diverse exploration of the hyperparameter space, mitigating overfitting and ensuring better generalization to unseen data. The method efficiently scales with large datasets, demonstrating adaptability to complex image classification problems in medical imaging.

Steps

Input: Local diabetic patient's health record

Output: Glaucoma chance or normal

Procedure

1. Incorporate patient dataset
2. Select all ensembles for optimization
3. Random search as optimizer
4. Execute the training model and obtain random search adaboost decision tree
5. Do performance analysis
6. Exporting the fit model to the workspace
7. Input new test data to make predictions
8. Result analyzed with experts.

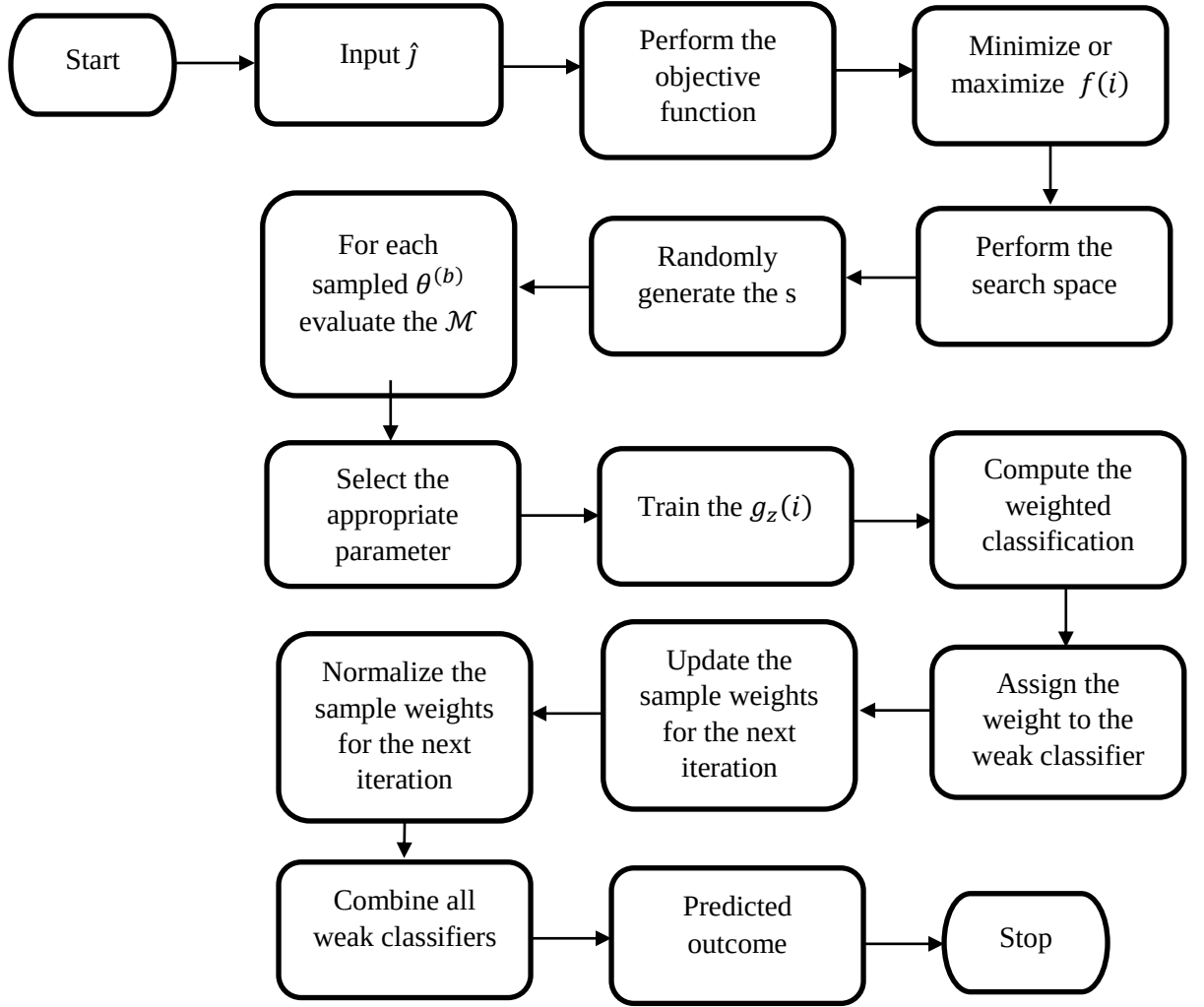


Figure 3.1 Flowchart Diagram of RSOAE

3.2.2 A Pragmatic Glaucoma Detection Based on DNN Strategy

The determination of this research is to cultivate a supervised learning procedure that incorporates retinal fundus characteristics and OCT records from the macular and optical disc region in order to accurately categories glaucoma disease and regular eyes as well as to assess its efficacy in diagnosing early glaucoma disease in clear manner. This research introduces a new DL strategy called DNCN (Figure 3.2), in which it associates the latest assessment metrics in order to identify the glaucoma disease in an efficient manner.

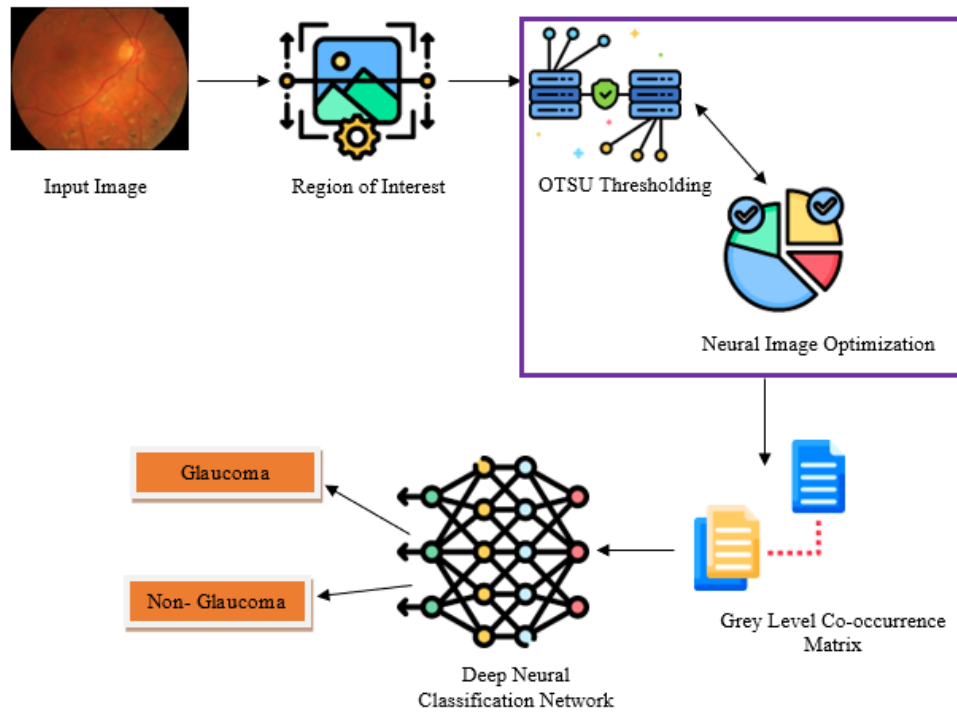


Figure 3.2 DNCN Architecture Diagram of the Glaucoma Disease Prediction

This proposed methodology of DNCN employs a number of strategies to accurately diagnose glaucoma at an initial stage (Figure 3.3). A new upgraded model is developed in this research to efficiently identify glaucoma disease through the process of digital image processing, which greatly aids in identifying glaucoma disease at an earlier stage through the use of a novel correlated multi-objective logic called Neural Image Optimization (NIO). The primary processes for diagnosing glaucoma include image cleaning, feature abstraction, image filtration and image classification, all of which are discussed in great depth below.

3.2.3 Neural Image Optimization (NIO) Method

The NIO method for glaucoma image segmentation refers to an advanced computational approach that utilizes neural networks to segment ROIs in glaucoma-related eye fundus images. This method leverages DL practices to advance the accuracy and efficacy of identifying structural variations in the ONH, RNFL, or other critical areas affected by glaucoma. It divides the image into

distinct regions, such as healthy tissue and glaucomatous regions, to aid in diagnostic and therapeutic decisions. NIO refines the segmented output using morphological operations or boundary smoothing techniques to ensure clinical relevance. This method is a cornerstone in computer-aided diagnosis systems, enabling early detection and monitoring of glaucoma progression. In equations (3.13) and (3.14), we normalize and filter the pixel intensity values of the glaucoma images, by using

$$X_{norm}(i, j) = \frac{X(i, j) - X_{min}}{X_{max} - X_{min}} \quad (3.13)$$

This equation normalizes the X of the $X(i, j)$ to a range [0 to 1]. It ensures consistent intensity distribution and reduces variations in illumination.

$$X_{filtrd}(i, j) = Med\{X(i + x, j + y) | x, y \in \{-k, \dots, k\}\} \quad (3.14)$$

Let us assume, x, y as the intensity of the image, and k as size of the median filter, it is used to decrease noise while conserving edges. This equation enhances the clarity of glaucoma-specific features like OD and cup boundaries. We performing the region-based segmentation to find the OD and OC boundaries and also compute the U-Net model through equations (3.15) and (3.16),

$$M_{contour} = \int_{\Omega} \alpha \|\nabla D(q)\|^2 + \beta \|\nabla^2 D(q)\|^2 + \gamma X(D(q)) ds \quad (3.15)$$

Let us assume, $M_{contour}$ as the Active Contour Model (ACM), α, β as the internal smoothness, and γ as the external image forces. In this equation a weighted energy functional that combines α, β and γ .

$$\hat{Q}(i, j) = f(X(i, j); \theta) \quad (3.16)$$

Let assume, $\hat{Q}$ as the segmentation maps, and θ as the learnable parameters. This equation is used to generate the $\hat{Q}(i, j)$ by the use of input image and θ . It is used to delineates glaucoma-affected regions in the image. After delineates the glaucoma pretentious regions in the image, we perform the deep feature embedding through equations (3.17) and (3.18). In equation (3.17), we execute convolutional layer, as

$$H_{k+1}(i, j) = \sigma(\sum_{x,y} C_{xy} H_k(i + x, j + y) + b) \quad (3.17)$$

Let us assume, H as the feature map, k as layer, C as the convolution kernel, b as bias term, and σ as the activation function. This equation extracts hierarchical features representing glaucoma-related patterns pooling via equation (3.18).

$$Q(i, j) = \max\{H(i + x, j + y) | x, y \in \{0, \dots, n - 1\}\} \quad (3.18)$$

It applies max-pooling over a window of size $n \times n$. This equation reduces spatial dimensions while retaining significant features. After retaining significant features, we perform back-propagation using the equation (3.19),

$$\mathcal{L}_D = 1 - \frac{2 \sum_{x=1}^N \hat{Q}_x Q_x}{\sum_{x=1}^N \hat{Q}_x^2 + \sum_{x=1}^N Q_x^2} \quad (3.19)$$

Let us assume, D as the Dice Similarity Coefficient (DSC) loss, $\mathcal{L}$ as loss function, $\hat{S}$ as the predicted segmentation, and S as ground truth. It is used to compute the overlap among $\hat{S}$ and S . In this equation we optimize the segmentation accuracy of glaucoma-affected regions. Then we perform the gradient descent using equation (3.20) for update the parameter,

$$\theta^{t+1} = \theta^t - \eta \nabla_{\theta} \mathcal{L} \quad (3.20)$$

Let us assume, θ as the model parameters, η as learning rate, and $\nabla_{\theta} \mathcal{L}$ as loss gradient. This equation minimizes the loss function to improve model performance. NIO enhanced accuracy in segmenting critical features of glaucoma, such as the OC and OD, compared to traditional methods. By leveraging optimized neural architectures, these methods minimize errors in detecting subtle structural changes associated with glaucoma, leading to better clinical outcomes. The approach enables automated and efficient segmentation processes, significantly reducing the time and determination mandatory for manual comment. It adapts to diverse datasets and imaging modalities, making it scalable for large-scale clinical applications.

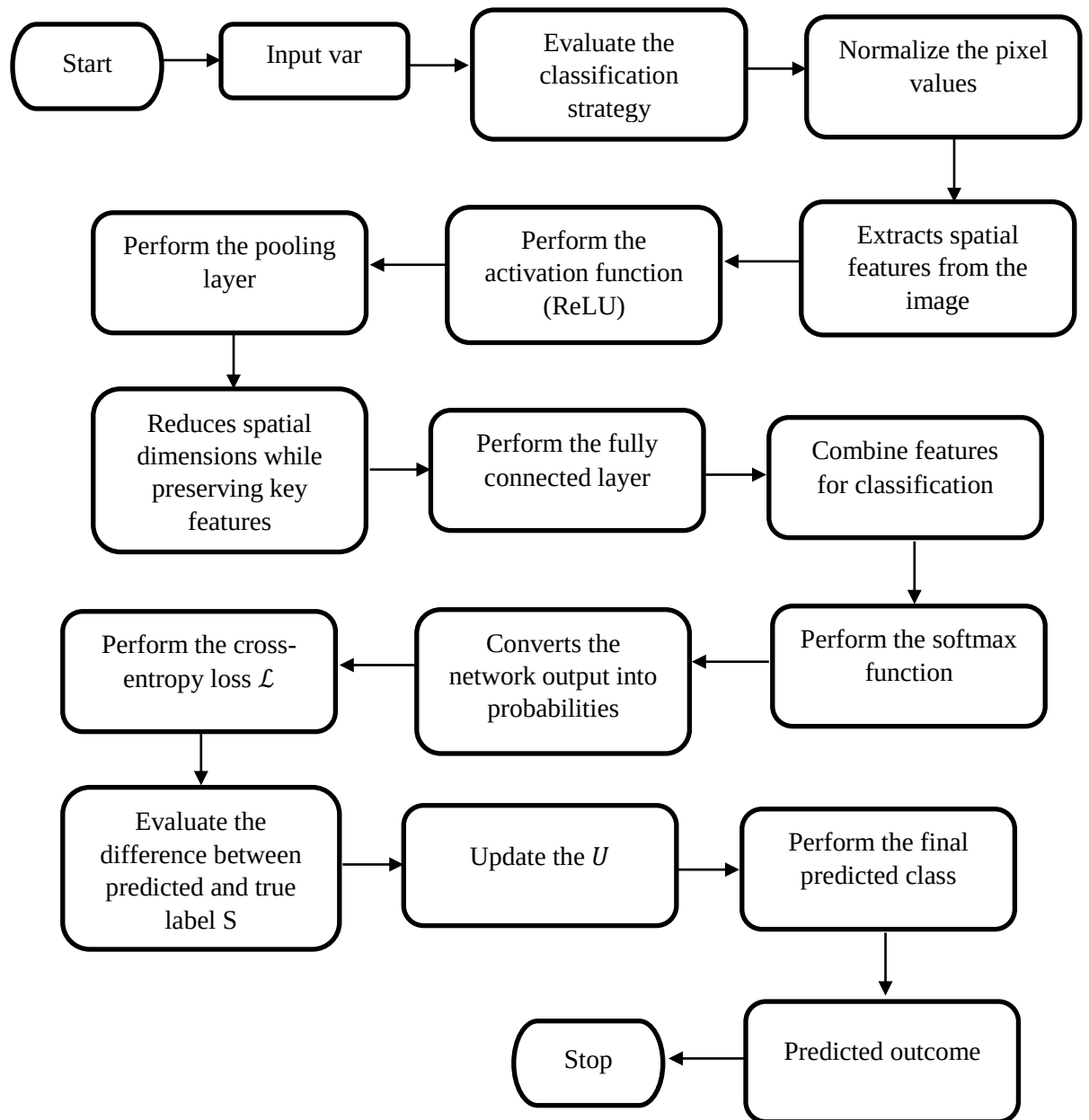


Figure 3.3 Flowchart Diagram of the DNCN Method

3.3 RESULTS AND DISCUSSION

The baseline attributes are preferred as per the experts. An individual with long years of diabetic mellitus, age, more CDR, and high IoP had a risk of glaucoma. Other correlated factors are blood sugar, visual acuity, family history, and red blood cell.

Table 3.1 Performance Accuracy of Training Classifiers

Classifier learners	Classifiers accuracy (%)
RSOAE	97.5
Bagged Tree	96.5
Boosted Tree	52.0
BOSVM	96.6

Table 3.1 depicts a comparison of the three ensemble tree models and BOSVM. Dataset is trained with the bagged tree. The boosted RSOAE machine learners are recapitulated based on accuracy and misclassification error rate. From this we conclude, RSOAE is the best fit model classifier for predicting glaucoma in the diabetic community.

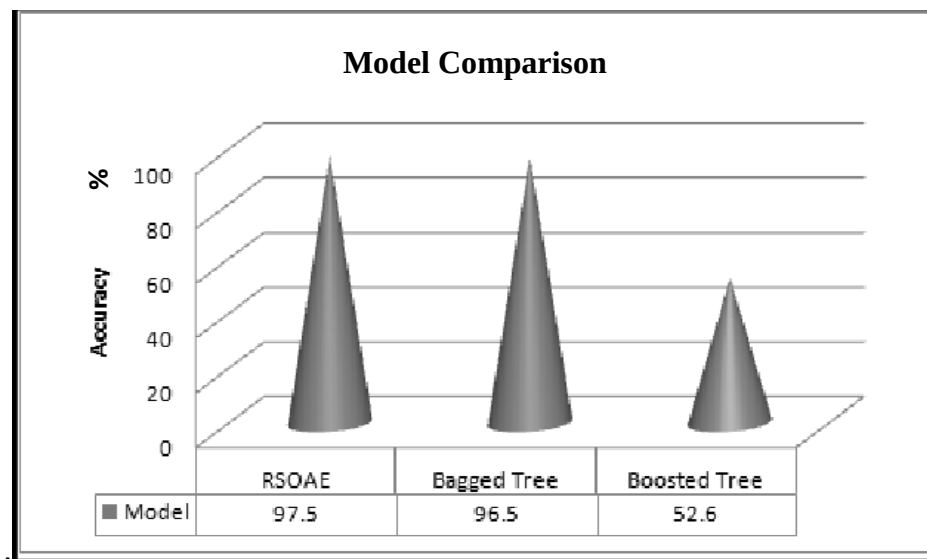
**Figure 3.4** Best Fit Model Selection

Figure 3.4 illustrates the details of the training model evaluation with 0.95 as a confidence level. The confusion matrix of the training dataset has been an estimated error rate of 2%, with a class rate of glaucoma chance as 0.019. From this, concluding the RSOAE is the best fit model for classification prediction of diabetic glaucoma with ROC is 0.97 on a false positive rate of 0.010 and true positive rate of 0.96154 (Figure 3.5).

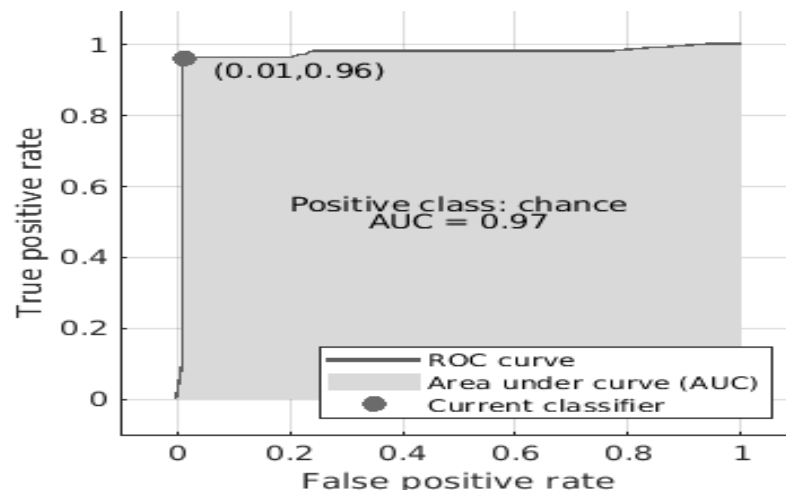


Figure 3.5 AUC of RSOAE Model

RSOAE had been evaluated by using twenty-five diabetic patients in real-time to predict glaucoma. This testing set has been attained an precision of 98.6% and AUC of 0.97. Figure 3.6 depicts the comparison of RSOAE and BOSVM performances.

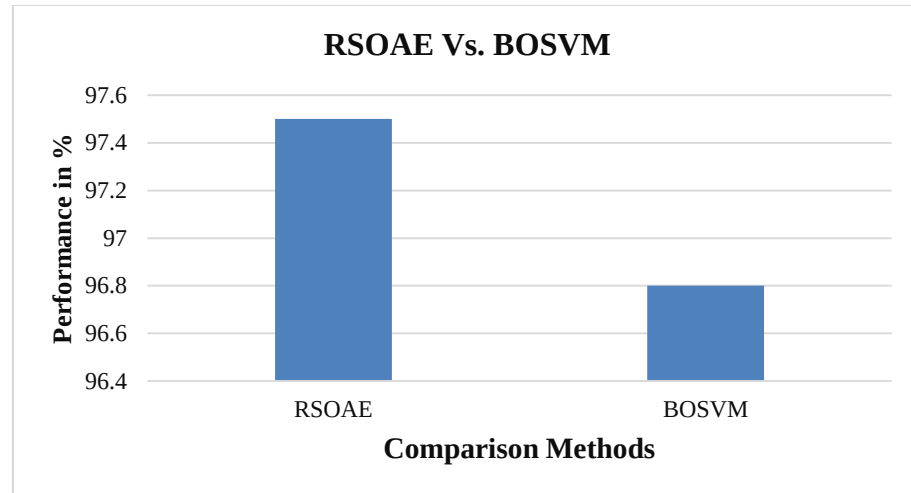


Figure 3.6 Comparisons of RSOAE and BOSVM

The test data provided to the RSOAE exemplary predicted the probability of disease, and the result sent to the expert's opinion.

Here, a few samplings of predictions are discussed to find the efficiency of the fit model with the specialists. The patient_id 10 had very little glaucoma because of no diabetes, average cdr, regular blood sugar, and other features. Experts also confirmed the same. Likewise, patient_id 20 had no risk and also approved by the Optometrists. But the patient id 15 had more risk on glaucoma and instant attention required. This was also confirmed by an Ophthalmologist.

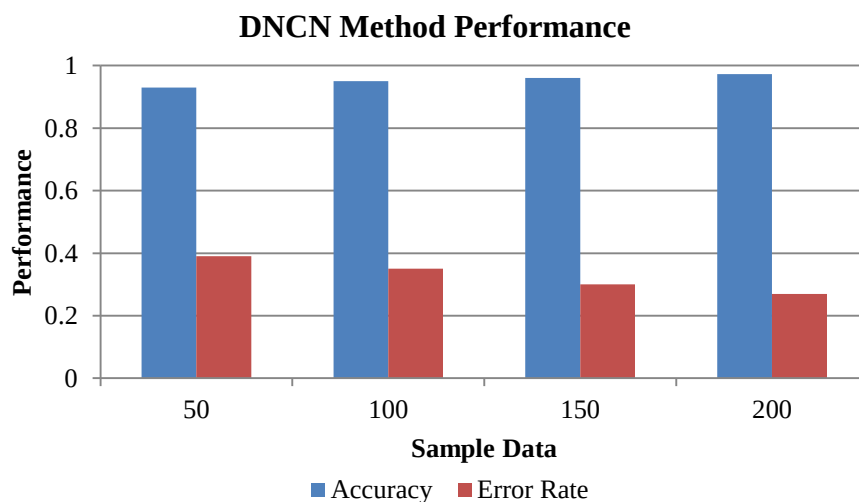


Figure 3.7 Analysis of Proposed Method Performance

Figure 3.7 describes the analysis of proposed DNCN method performance of accuracy and error rate. Here the samples of 50, 100, 150 and 200 numbers of images in different proportions are occupied into concern to train the datatable accordingly. The proposed DNCN method attained accuracy is 0.973 and error rate is 0.27.

3.4 SUMMARY

At this high point, the ML classifier played a crucial part in screening of glaucoma at the earlier stage, to avoid visual impairment. Several machine learners are used to formulating the fit model to predict glaucoma and preprocessed to overcome this. Tailored dataset was used for this evaluation under three different ensembles and compared with BOSVM learner. From this, RSOAE is the best fit ensemble predictor, which obtained an accuracy of 97.5% and 98.01% in both the phases of training and validation. In the upcoming process, this model has been incorporated with a customized optimizer to rise the accuracy of the prediction. This model has been pertinent to link patient's e-health database with clinical people to make the decision swiftly. This fresh clinical support system can forecast post-fungal eye infection. An effective strategy for detecting glaucoma disease in earlier stage as well as the identification of the progression of glaucoma disease is crucial for effective treatment.

The DNCN based classification logic is aimed to classify retinal images and OCT as glaucoma, non-glaucoma or suspicious. This is accomplished by integrating structural and functional components from the fundus with the thickness of the retinal fibre layer computed from OCT image. Glaucoma disease detection effectiveness can be found by using the proposed approach of DNCN with the accuracy ratio of 0.973 with the least error estimation of 0.27%.